Phase 1 Project Proposal

Publisher-Invariant Fake News Detection under Cross-Source Distribution Shift

Group ID: F26-08

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1. Problem Statement:

Most fake news detection models are trained and tested on articles from the same set of news outlets. Because of this, a model can score well just by picking up the habits of a particular publisher which generally leans towards the boundary of overfitting, such as its writing style, its formatting, its headline wording, or even the name of the website which is not a thing to happen. If a publisher happens to have more false stories in the training data, the model starts linking that publisher with the label "fake" instead of learning anything about the content. When the same model is then tested on articles from an outlet it has never seen, its accuracy usually drops a lot, this just shows that the model is memorizing or is over trained on some publisher's or training data than it gives answers on test data being biased and doesn't perform as it is supposed to.

This project looks at that problem and tries to solve it. We want to build a text based fake news classifier that keeps its performance when the source of the news changes, so that it relies on the claim and how it is written and not on who published it. We will first measure how big the drop is when a standard model is tested on unseen publishers and then try to reduce that drop using publisher-invariant training methods.

2. Dataset:

We will be using the FakeNewsNet, a public collection of news articles labelled as real or fake. The articles come from two fact-checking sources, PolitiFact (political news) and GossipCop (entertainment news basically what we call gossip as the name suggests). For each article the repository gives the news content (title, body text, source URL and images where available) along with social context such as tweets and user information which are all useful in telling whether some information is true or not.

Dataset link: https://github.com/KaiDMML/FakeNewsNet/tree/master/dataset

The dataset is multimodal, but mainly of the text (NLP) type. Our main experiments will use the article title and body text. The publisher of each article will be taken from the domain in its

URL and will only be used to make the train/test splits and to apply the invariance methods. It will not be given to the classifier as an input. The repository provides article links and a download script instead of the full text, so part of the work will be crawling and cleaning the articles, and some dead links will have to be dropped.

3. Scope of the Project:

Included:

- Collecting and cleaning the FakeNewsNet articles, removing duplicates and articles with missing or very short text, and extracting the publisher from each URL(only if we think the URL is too short and dont have any endpoint, domain, path, query,etc).
- Two evaluation setups which we will be doing are: a normal random split, and a publisher-held-out split where the test publishers never appear in training(as it would be a great way to check the actual prediction of the model on the new data on which it hasn't been trained).
- Baseline classifiers for this would be TF-IDF features with logistic regression, and a fine-tuned pretrained transformer such as BERT or RoBERTa.
- One or two methods to reduce dependence on the publisher, for example masking publisher names and boilerplate text, and adversarial training that penalizes the model when the publisher can be predicted from its features, we will try to do it this way so that the model doesn't too much rely on the source of the news.
- Comparing the models on accuracy, precision, recall and F1, and reporting how much each one loses when moving from the random split to the unseen-publisher split and the reporting of the evaluations would be done in full honesty with actual evaluation values.
- A short error analysis of which articles and publishers are hardest, the model will make or it may mistakes by predicting an article wrong if it does it then we will find that article, group it by other such articles and then will try to find why it is doing so like this.

Not included:

- Modelling how news spreads on social media, such as retweet networks and user graphs.
- Image-based detection. Images may be looked at if time allows, but they are not a committed part of the project.
- Checking claims against outside knowledge sources or building a live fact-checking system.
- Deploying the model as a website or real-time tool, and languages other than English.

Expected outcome:

A working pipeline and a comparison showing how standard fake news classifiers behave on unseen publishers, and whether the methods we try give a measurable improvement in that setting. We will report the results honestly, including any cases where the methods do not help.